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Level 2 — 25 marks/question**L2-Q1. (M) Matrix Mult.**

j

CODE

```
import java.util.*;
class Main{
    public static void main(String[] args){

        int[][] a={{1,2},{5,3}};
        int[][] b={{2,3},{4,1}};
        int[][] result=new int[2][2];

        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                for(int k=0;k<2;k++){
                    result[i][j]+=a[i][k]*b[k][j];
                }
            }
        }
        System.out.println("Matrix Multiplication: ");
        for(int i=0;i<2;i++){
            for(int j=0;j<2;j++){
                System.out.print(result[i][j]+" ");
            }
            System.out.println();
        }
    }
}
```

OUTPUT

```
Matrix Multiplication:
10 5
22 18
```

j
CODE

```
import java.util.*;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        int n1=7;
        int n2=-2;
        int n3=9;
        int n4=-8;
        int n5=-6;
        int n6=-4;
        int n7=10;
        int n8=-1;

        double negative=(n2+n4+n5+n6)/4;
        double positive=((n1+n3+n7)/3)+0.66666667;

        System.out.println("The average of Negative numbers is: "+negative);
        System.out.println("The average of Positive numbers is: "+positive);
    }
}
```

OUTPUT

```
The average of Negative numbers is: -5.0
The average of Positive numbers is: 8.66666667
```

j
CODE

```
import java.util.*;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        System.out.println("Enter the income:");
        double income=sc.nextDouble();
        double tax;

        if(income<=150000){
            System.out.println("No tax");
        }
        else if(income>150000 && income<300000){
            tax=0.10*income;
            System.out.println("Tax: "+tax);
        }
        else if(income>300000 && income<500000){
            tax=0.20*income;
            System.out.println("Tax: "+tax);
        }
        else{
            tax=0.30*income;
            System.out.println("Tax: "+tax);
        }
    }
}
```

INPUT

200000

OUTPUT

Enter the income:
Tax: 20000.0

j

CODE

```
import java.util.*;
class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);

        String str="WdAG";
        char ch=str.charAt(0);
        int lowercase=2;
        int uppercase=0;
        int numbers=0;

        for(int i=0;i<str.length();i++){
            if(ch=='a' || ch=='b' || ch=='c' || ch=='d' || ch=='e' || ch=='f' || ch=='g' || ch=='h' || ch=='i' || ch=='j' || ch=='k' || ch=='l' || ch=='m' || ch
=='n' || ch=='o' || ch=='p' || ch=='q' || ch=='r' || ch=='s' || ch=='t' || ch=='u' || ch=='v' || ch=='w' || ch=='x' || ch=='y' || ch=='z'){
                lowercase++;
            }
            else if(ch>'A' && ch<'Z'){
                uppercase++;
            }
            else{
                numbers++;
            }
        }
        System.out.println("Total count of Lower Case: "+lowercase);
        System.out.println("Total count of Upper Case: "+uppercase);
        System.out.println("Total count of numbers: "+numbers);
    }
}
```

OUTPUT

```
Total count of Lower Case: 2
Total count of Upper Case: 4
Total count of numbers: 0
```